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Series tests for convergence

Determine if the following series converge or diverge. If applicable, determine if the series are absolutely convergent, conditionally convergent, or divergent (this is applicable if you have some $a_n < 0$. If possible, determine what the series converges to.)

Some hints and brief methods for you to check your work. These are not complete solutions.

1. $\sum_{m=1}^{\infty} \frac{1}{m^2}$

This series converges by p-test.

2. $\sum_{n=1}^{\infty} \frac{1}{2^n}$

This is a geometric series, and it converges to 1.

3. $\sum_{q=1}^{\infty} (-3)^q$

Series diverges because of either divergence test, ratio test, geometric series test, or root test.

4. $\sum_{n=1}^{\infty} \ln\left(\frac{n}{3n+1}\right)$

Series diverges because of divergence test.

5. $\sum_{k=1}^{\infty} \frac{k^{2k}}{(1+2k^2)^k}$

By root test, the series is absolutely convergent.

6. $\sum_{m=1}^{\infty} \frac{m^3}{5^m}$

By ratio test, the series is absolutely convergent.

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7. $\sum_{j=1}^{\infty}(-1)^j\frac{1}{j+2}$

Using alternating series test and a comparison test, we can show this series is conditionally convergent.

8. $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{\sqrt{n}}{n+1}$

Using alternating series test and a comparison test, we can show this series is conditionally convergent.

9. $\sum_{r=1}^{\infty}(-1)^{r+1}$

This series diverges by divergence test or geometric series test.

10. $\sum_{b=1}^{\infty}\frac{b}{b^3+5}$

This series converges, can show with a comparison test (Direct works on this one, you might try that for practice, but limit should also work)

11. $\sum_{k=1}^{\infty}\frac{k^3-k+1}{\sqrt[3]{k^9+k^5}}$

Using limit comparison test, this series diverges. You may find divergence test useful here too.

12. $\sum_{j=1}^{\infty}\frac{\cos(3j)}{(1+1.2)^j}$

This series is absolutely convergent. We can compare $\sum_{j=1}^{\infty}\frac{|\cos(3j)|}{(1+1.2)^j}$ to $\sum_{j=1}^{\infty}\frac{1}{(1+1.2)^j}$ to discuss its convergence.

13. $\sum_{n=2}^{\infty}\frac{1}{n(\ln(n))^2}$

By integral comparison test, this series converges.

14. $\sum_{k=1}^{\infty}\frac{(-1)^k(k+1)3^k}{(k+2)!}$

By ratio test, this series is absolutely convergent.